

Signal Processing: Some useful trigonometric identities (IDs)

Double angles:

$$\sin(2\theta) = 2 \sin(\theta) \cos(\theta) \quad (1)$$

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) \quad (2)$$

Power reduction:

$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2} \quad (3)$$

$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2} \quad (4)$$

$$\sin^2(\theta) \cos^2(\theta) = \frac{1 - \cos(4\theta)}{8} \quad (5)$$

Products-to-sums:

$$\cos(\theta) \cos(\phi) = \frac{\cos(\theta - \phi) + \cos(\theta + \phi)}{2} \quad (6)$$

$$\sin(\theta) \sin(\phi) = \frac{\cos(\theta - \phi) - \cos(\theta + \phi)}{2} \quad (7)$$

$$\sin(\theta) \cos(\phi) = \frac{\sin(\theta + \phi) + \sin(\theta - \phi)}{2} \quad (8)$$

$$\cos(\theta) \sin(\phi) = \frac{\sin(\theta + \phi) - \sin(\theta - \phi)}{2} \quad (9)$$

Useful integral identities:

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \quad (10)$$

$$\int_{-\infty}^{\infty} e^{-ax^2} \cos(2\pi kx) dx = \sqrt{\frac{\pi}{a}} e^{-\frac{\pi^2 k^2}{a}} \quad (11)$$